CSE 318

Solve n-Puzzle using A* search:

priority function, $f(n) = g(n) + h(n)$,

$g(n) \rightarrow$ the actual cost from initial state to nth state

$h(n) \rightarrow$ estimated cost from node n to the goal state.

this is called a heuristic

(i) Hamming distance: The numbers of blocks in the wrong position.

7	2	4
6		5
8	3	1

initial board

1	2	3
4	5	6
7	8	

goal board

hamming distance: 7

Digits	1	2	3	4	5	6	7	8
Is placed correct?	1	0	1	1	1	1	1	1

(ii) Manhatan distance: The sum of the Manhattan distances (sum of vertical and horizontal distance) of the blocks to their goal positions.

CS CamScanner

Digits	1	2	3	4	5	6	7	8
Row dist.	2	0	2	1	0	0	2	0
Column dist	2	0	1	2	1	2	0	1

total = 2 + 2 + 2 + 1 + 1 + 2 + 1 + 2 + 2 + 1
= 16

(3) Euclidean heuristic: it calculates the straight-line distance between a tile's current position and its goal position.

current state:

1	2	3
4	5	6
0	7	8

goal state:

1	2	3
4	5	6
7	8	0

for tile 7: current position: (2,1) ; Goal position: (2,0)

~~for ti~~ distance = $\sqrt{(2-2)^2+(1-0)^2} = 1$

for tile 8: current position: (2,2) ; Goal. position: (2,1)

distance = $\sqrt{(2-2)^2+(2-1)^2} = 1$

∴ total euclidean heuristic is: 1+1 = 2

Inversion Count:-

A pair of tiles form an inversion if the values of tiles are in reverse order of their apperance in goal state.

goal state: 1 2 3 5 4 7 8 6 -

current state: 2 3 4 7 6 8 1 5 -

inversion count for: 2 → 1, 3 → 1, 4 → 2, 7 → 2, 6 → 3, 8 → 2
1 → 0, 5 → 0 total: 13

Solvability:

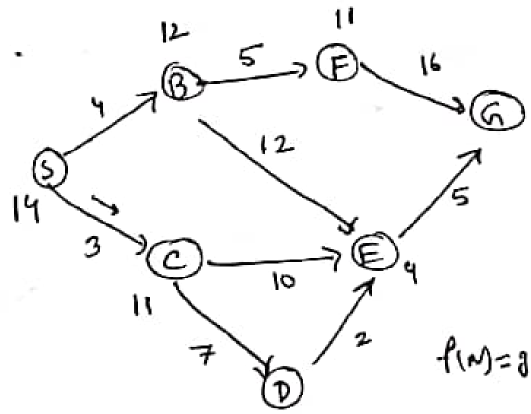
(1) if grid size k is odd, then No. of inversions should be even number.

(2) if grid size k is even, then

(a) the blank is on an even row counting from the bottom and number of inversion is odd.

(b) the blank is on an odd row counting from the bottom and number of inversion is even.

Simulations



$f(n) = g(n) + h(n)$

Start state: S , Goal state: G

S→B : 4+12 = 16 ✓✓

S→C : 3+11 = 14 ✓

S→C→D : 3+7+6 = 16 ✓✓

S→C→E : 3+10+4 = 17

S→B→E : 4+12+4 = 20

S→B→F : 4+5+11 = 20

S→C→D→E : 3+7+2+4 = 16 ✓✓✓

S→C→D→E→G :

3+7+2+5 = 17

Time complexity : $O(b^d)$

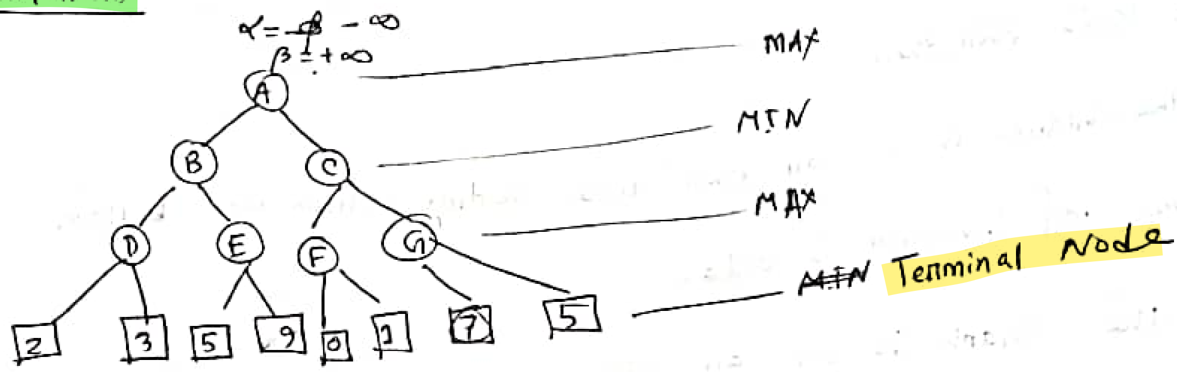
b = branch factor

d = depth

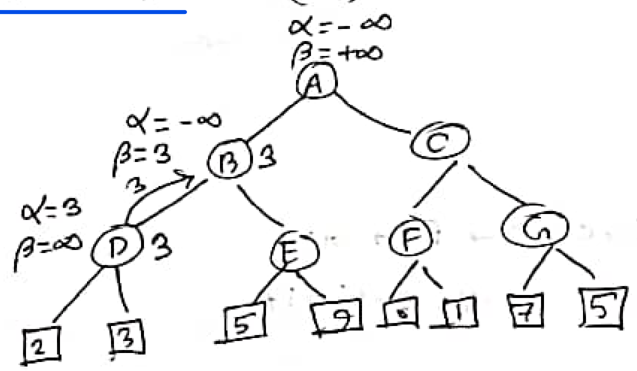
Space complexity : $O(b^d)$

Adversarial Search :

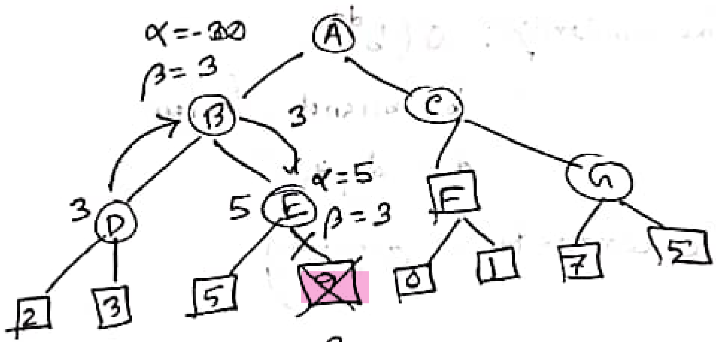
Simulations



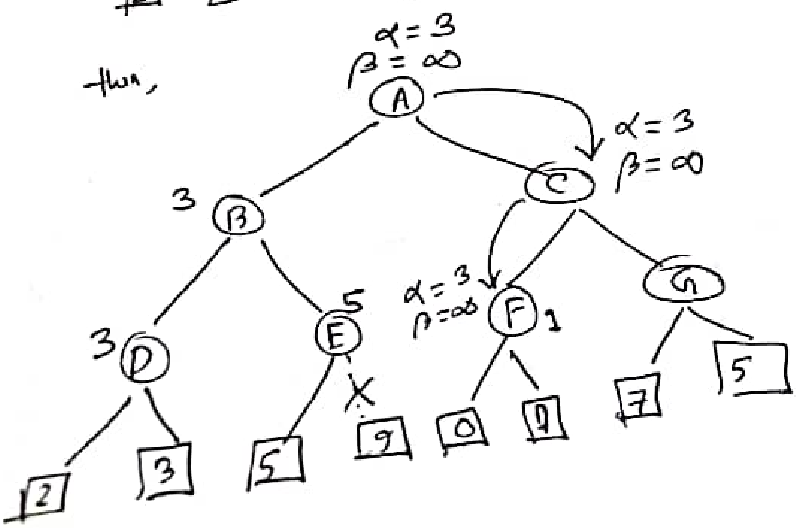
for node D : max(2, 3) = 3

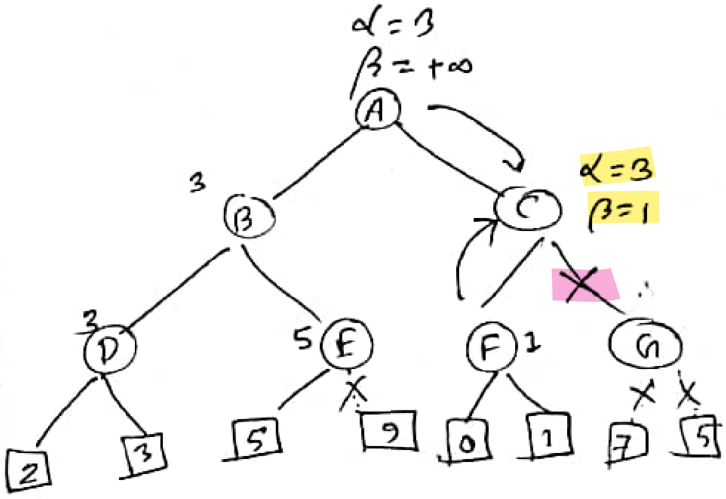


then,



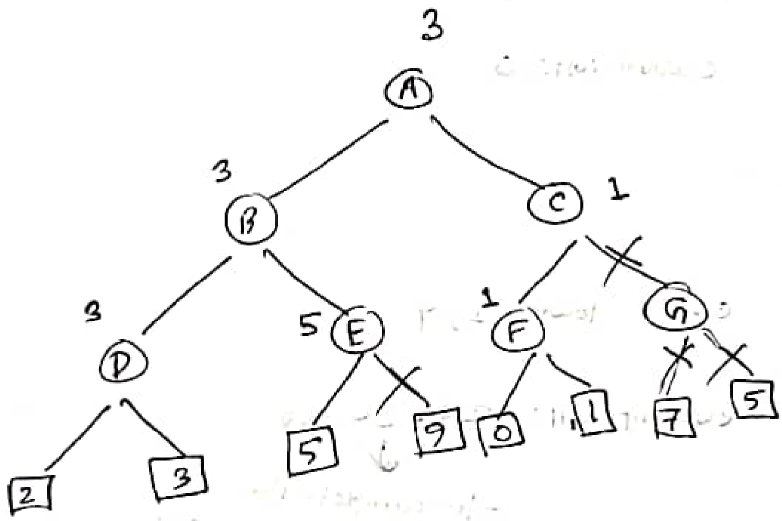
then,





as, $\boxed{\alpha \geq \beta}$

Now C returns the value of 1 to A here the best value for A is $\max\{3,1\} = 3$



Worst ordering: Doesn't prune any of the leaves of the tree.

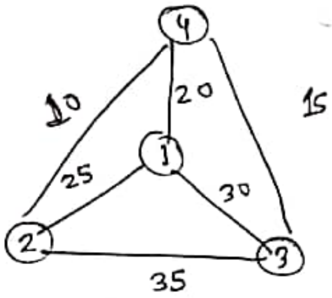
time complexity: $O(b^m)$

Ideal ordering: best moves occur ~~at~~ the left side of the tree.

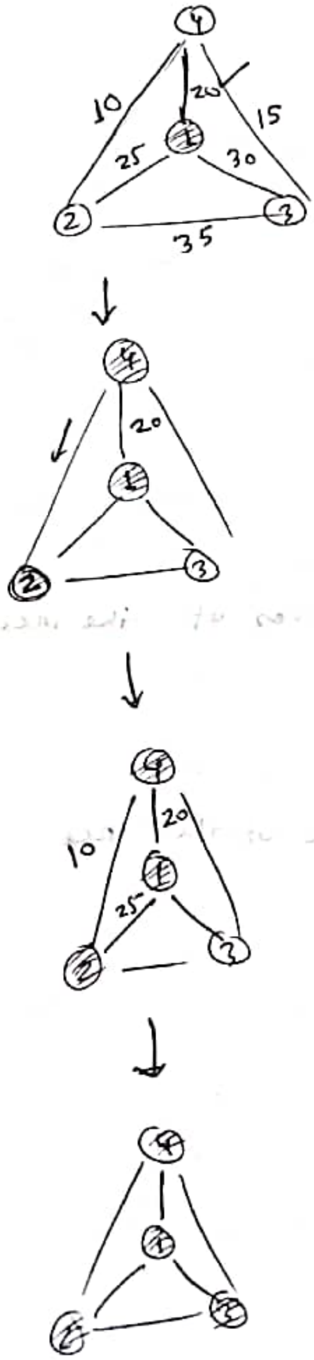
time complexity: $O(b^{\frac{m}{2}})$

CS CamScanner

Solving TSP by local Search:



1. Nearest Neighbour (NN):



current tour: 1

current cost: 0

current tour: 1, 4

current cost: 20 + 20 = 40

to complete the tour come back to initial state

current tour: 1, 4, 2

current cost: 20 + 10 + 25 = 55

current tour: 1, 4, 2, 3

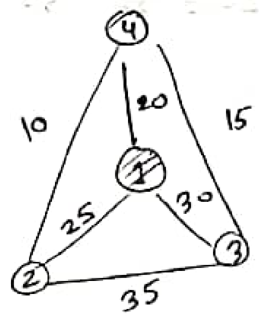
current cost: 20 + 10 + 35 + 30 = 95

CamScanner

Pros:
1. Simple, Intuitive and relatively efficient.
2. Empirically OK, esp. on Euclidean TSP

Cons.
1. Greedy: can easily miss shortcut paths.

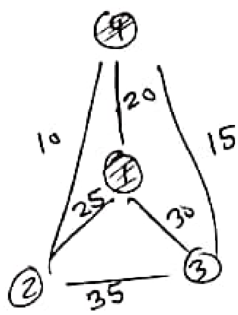
Nearest Insertion (NI):



cur. tour: 1

cost: 0

↓



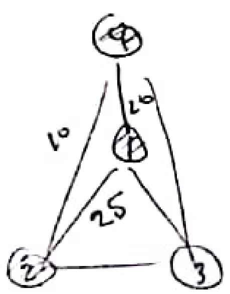
cur. tour: 1, 4

cost: $20+20=40$

Next vertex: 2

After 1: $w(1,2)+w(2,4)-w(1,4) = 25+10-20 = 15$

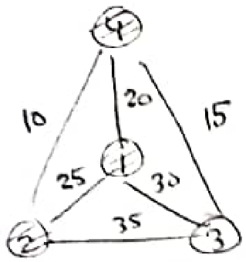
After 4: $w(4,2)+w(2,1)-w(1,4) = 10+25-20 = 15$



curr. tour: 1, 2, 4

curr. cost: $25+10+20=55$

Next vertex: 3



cur. tour: 1, 2, 4

cost: $25 + 10 + 20 = 55$

After: 1 $\quad \omega(1,3) + \omega(3,2) - \omega(1,2) = 30 + 35 - 25 = 40$

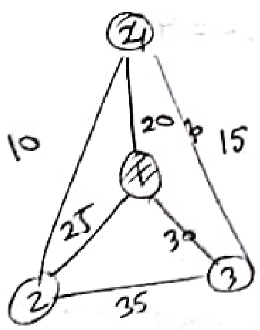
After: 2 $\quad \omega(2,3) + \omega(3,4) - \omega(2,4) = 35 + 15 - 10 = 40$

After: 4 $\quad \omega(4,3) + \omega(3,1) - \omega(4,1) = 15 + 30 - 20 = \cancel{55}\ 35$

∴ Tour: 1, 2, 4, 3

cost: $25 + 10 + 15 + 30 = 80$

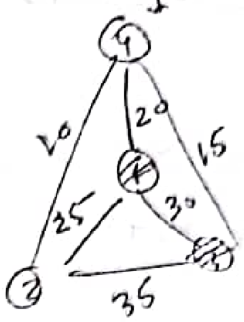
Farthest Insertion (FI) :



current tour: 1

cur. cost: 0

taking 3



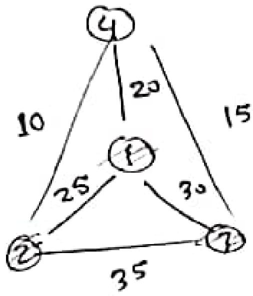
curr. tour: 1, 3

cur. cost: $30 + 30 = 60$

taking 2

After: 1 $\omega(1,2) + \omega(2,3) - \omega(1,3) = 25 + 35 - 30 = 30$

After: 3 $\omega(3,2) + \omega(2,1) - \omega(1,3) = 35 + 25 - 30 = 30$



curr. tour: 1,2,3

cost: $25 + 35 + 30 = 90$

taking 4

After 1 $\omega(1,4) + \omega(4,2) - \omega(1,2) = 20 + 10 - 25 = 5$

After: 2 $\omega(2,4) + \omega(4,3) - \omega(2,3) = 10 + 15 - 35 = -10$

After: 3 $\omega(3,4) + \omega(4,1) - \omega(3,1) = 15 + \underline{20} - 30 = +5$

tour: 1, 2, 4, 3

cost: $25 + 10 + 15 + 30 = 80$

Decision Tree :

Given dataset.

high, low, medium

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

Q. What is the entropy H(passed) ?

soln:

$$H(passed) = -\left(\frac{2}{6}\log_2\frac{2}{6} + \frac{4}{6}\log_2\frac{4}{6}\right)$$

$$= 0.92$$